

# Michael Plessner

[plessner\(at\)psu.edu](mailto:plessner(at)psu.edu)  
[michaelplessner.github.io](https://github.com/michaelplessner)

## Education

---

|                                                                            |                    |
|----------------------------------------------------------------------------|--------------------|
| <b>Pennsylvania State University (PSU) — PhD in Physics</b>                | <b>2022 - 2027</b> |
| <i>John Randall Shuman Troxell Memorial Scholarship in Physics</i>         | 2025               |
| <i>Edward A. and Rosemary A. Mebus Graduate Fellowship in Physics</i>      | 2024               |
| <i>Edward M. Frymoyer Honors Scholarship</i>                               | 2024               |
| <b>Pennsylvania State University (PSU) — MSc in Physics</b>                | <b>2020 - 2022</b> |
| <i>Paul Berg and Daniel J. Larson distinguished graduate fellowship</i>    | 2020               |
| <b>The University of Hong Kong (HKU) — Semester Abroad</b>                 | <b>2016</b>        |
| <b>Northeastern University (NEU) — BSc in Physics with Honors</b>          | <b>2015 - 2019</b> |
| <i>Merit scholarship for four years, also awarded a Dean's scholarship</i> |                    |
| <i>Presidential global scholarship to perform research in Italy</i>        |                    |
| <i>Presidential global scholarship to study in Hong Kong</i>               |                    |

## Research

---

**PSU — PhD Supervisor: Dr Jacob Bourjaily** 2022 - 2027

Broadly interested in the mathematical structures of QFTs and improving “on-shell” methods.

Ongoing research programs include,

*Colour in gauge theory, applications of classical and quantum representation theory,  
Covariant formalisms for massive particles and generalized recursion relations,  
Mathematica packages for handling the math and physics of scattering,  
Generalized double copy and colour-kinematics duality,  
Combinatoric models of amplitudes.*

**NEU — Supervisor: Dr James Halverson** 2019 - 2020

Studied solvability of Diophantines in string index calculations under topological transitions.

**CERN / CMS / NEU — Supervisor: Dr Toyoko Orimoto** 2018 - 2019

Used ROOT to perform test beam timing resolution analyses on HL-LHC ECAL hardware.

**Università degli Studi di Milano — Supervisor: Dr Marcel Di Vece** 2017 - 2018

Modeled “planetary” nanoparticle systems formed by a sputtering cluster source in Python.

**CERN / ATLAS / HKU — Supervisor: Dr Yanjun Tu** 2016 - 2017

Used ROOT to simulate backgrounds and search for SUSY in dilepton events at CERN.

## Publications and Preprints

1. S. Caron-Huot, J. Fei, **M. Plesser**, *Canonical Bases for Classical Group Tensor Products*, In Preparation
2. J. Bourjaily, **M. Plesser**, *Covariant BCFW for all Masses*, [2610.10686](#)
3. J. Bourjaily, **M. Plesser**, P. Velie, *Constructing Concrete Colour Tensors for Simple Lie Algebras*, [2609.13365](#)
4. J. Bourjaily, **M. Plesser**, P. Velie, *Tree Amplitudes in Pure Gauge Theory with Charged Matter*, [2604.18672](#)
5. J. Bourjaily, **M. Plesser**, C. Vergu, *The Colour Dependence of Amplitudes*, **JHEP** 08 (2026) – [2512.23806](#)
6. J. Bourjaily, **M. Plesser**, C. Vergu, *The Many Colours of Amplitudes*, [2412.21189](#)
7. J. Bourjaily, **M. Plesser**, et al, *Gauge-Invariant Double-Copies via Recursion*, **Phys. Rev. Lett.** 131 (2023) – [2307.02542](#)
8. J. Halverson, **M. Plesser**, F. Ruehle, J. Tian, *Kahler Moduli Stabilization and the Propagation of Decidability*, **Phys. Rev. D** 101 (2020) – [1911.07835](#)
9. S. Marom, **M. Plesser**, et al, *'Planetary' silver nanoparticles originating from a magnetron sputter plasma*, **J. Phys. D: Appl. Phys** 52 (2019)

## Teaching Assistantships at Penn State University

|                                                           |                  |
|-----------------------------------------------------------|------------------|
| <b>Introductory Mechanics (PHYS 211)</b>                  | 2024, 2026       |
| <b>Introductory Electromagnetism (PHYS 212)</b>           | 2022, 2023, 2024 |
| <b>Thermal Physics (PHYS 213)</b>                         | 2022             |
| <b>Wave Physics (PHYS 214)</b>                            | 2022, 2025       |
| <b>Algebra-Based Introduction to Mechanics (PHYS 250)</b> | 2025             |
| <b>Electromagnetism (PHYS 400)</b>                        | 2021             |
| <b>Subatomic Physics (PHYS 406)</b>                       | 2022, 2025       |
| <b>Quantum Mechanics I (PHYS 410)</b>                     | 2021, 2025       |
| <b>Graduate Quantum Mechanics II (PHYS 562)</b>           | 2022             |
| <b>Computational Resources and Curriculum Development</b> | 2026             |

## Invited Talks and Visits

**Invited talk at Penn State's Math department** (upcoming)

*"Canonical bases for quantum groups and applications to physics"*

**Invited to visit McGill University by Simon Caron-Huot** McGill, QC, CA, 2026

**Invited to visit the IAS by Jacob Bourjaily** IAS, NJ, USA, 2026

**Invited to visit the IAS by Nick Early and Nima Arkani-Hamed** IAS, NJ, USA, 2026

**Invited talk at UCT Applied Maths department** UCT, Cape Town, SA, 2025

*"What can representation theory teach us about gauge theory?"*

**Invited talk at April APS Conference** Denver, CO, USA 2019

*"Test beam studies of precision timing with the upgraded CMS-ECAL"*

## Conferences, Summer Schools, and Workshops

**AI in CAmplitudes** UCLA, CA, USA 2026

**Loop-The-Loop 3 Workshop on Amplitudes** (Online), 2026

**Workshop at the IAS hosted by Nima Arkani-Hamed** IAS, NJ, USA, 2026

**Prospects in Theoretical Physics (PiTP) Summer School** IAS, NJ, USA, 2026

*Gong show: "QED from  $N=4$ "*

**Amplitudes Summer School** U. of Southampton, UK, 2026

**Symbology@15 Conference** MPP, Munich, DE, 2025

*Gong show: "Masses, flavours, and colours in amplitudes"*

**TASI Summer School** CU Boulder, CO, USA, 2025

*Gong show: "Colour structures and amplitudes"*

**Michigan Interdisciplinary Meeting on Amplitudes** U. of Michigan, MI, USA, 2025

*Gong show: "Graph games from representation theory"*

**Loop-The-Loop 2 Workshop on Amplitudes** (Online), 2025

**Strings and Geometry Conference** ICTP Trieste, IT, 2025

**Loop-The-Loop Workshop on Amplitudes** (Online), 2024

**Mathematical Aspects of Scattering Amplitudes Conference** GGI, Florence, IT, 2024

**Amplitudes Conference and Summer School** IAS, NJ, USA, 2024

*Gong show: "A ridiculous formula for the number of  $SU(N)$  gluonic colour factors"*

**Amplitudes Conference and Summer School** CERN, FR / CH, 2023

*Gong show: "Covariant BCFW for all masses"*

**Amplitudes Conference and Summer School** Charles University, Prague, CZ, 2022

**The Amplitude Games Summer School** MITP (Online), 2021

**Quantum Gravity Summer School** Bard College, 2019

## Outreach and Mentorship

---

- 1) Provided tutoring and career advice to minority students at the HBCU (historically black college / university) North Carolina Central University as part of the PREM (Partnership for Research and Education) program
  - Graduate Electromagnetism** 2024, 2025
  - Undergrad Electromagnetism and Quantum Mechanics** 2023, 2024, 2025
- 2) Peer supervisor for junior PhD students working in Dr. Bourjaily's group at Penn State
  - Philip Velie** 2023-2027
  - Malek Telfah** 2026-2027
- 3) Tutored Penn State physics PhD students remediating first-year courses 2025
- 4) Research talks at Penn State's "Primordial Universe and Gravity" (PUG) seminar
  - "TBD" (upcoming)** 2026
  - "Why (Non-Abelian) Gauge Theory?"** 2025
  - "Correcting Coulomb and Newton with On-Shell Methods"** 2024
  - "Quantum Field Theory Without Fear"** 2023
- 5) Research talk to the physics and chemistry club at State Area High School
  - "A Brief History of QFT and Why the Sky is Blue"** 2022
- 6) Research talk at Penn State's "Mathematical Aspects of Physics" (MAP) seminar
  - "Why Good Variables Matter: Representations of Poincaré"** 2022

## Rock Climbing

---

- Semi-professionally sponsored by Mad Rock Climbing from 2014 - present.
- Commercial Routesetter with over a decade of experience.
  - *Routes:*
    - Salathé Wall** on **El Capitan** (5.9 C2, 3500' Yosemite, USA),
    - Snake Dike** on **Half Dome** (5.7 R, 2000' Yosemite, USA),
    - Time Wave Zero** on **El Toro** (5.12a, 2300' El Potrero Chico, Mexico),
    - Flatirons scramble** during the TASI 2025 summer school in Boulder, CO
    - El Camino Royale** 500' - **Royal Arch** 100' - **Dodge Block** 700'
  - *Boulders:*
    - Mooiste Meisie** (v13 / 8B, South Africa),
    - Donkeystrength** (v12 / 8A+ FA, South Africa),
    - Freak Brothers** (v11 / 8A, Switzerland),
    - Monstrous Moonshine** (v11 / 8A FA, NC, USA),
    - 973 to the Dangerzone** (v6 / 7A R FA, Hong Kong)